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Preamble

The present *readme* file provides some information about the *Mult* and *Fill* assembler programs available in the [04](#) Bitbucket directory.

Mult.asm program

The code is based on a simple algorithm described in the [asm](#) source file.

Fill.asm program

The program fills the screen memory zone with the content of the [wordvalue](#) variable. Its content is set to 0 to clear the screen or to -1 (thus all bits set to 1) to blacken it. Remembering how to test it was not so easy and since we have to check visually if it runs well, here is how I do it on my Windows PC :

- Call the [CPUEmulator.bat](#) tool.
- *File / Load Program*
- Select [Fill.asm](#) and click *Load ROM*
- *Animate : No animation* and *View : Screen*
- Run (▶▶)
- Check that the keyboard button is selected.
- Maintain one key pressed to blacken the screen, release the key to clear it.

CPU Emulator (2.5) - C:\Perso\Work\nand2tetris\projects\04\fill\Fill.asm

File View Run Help

The screenshot shows the CPU Emulator (2.5) interface. The title bar reads "CPU Emulator (2.5) - C:\Perso\Work\nand2tetris\projects\04\fill\Fill.asm". The menu bar includes "File", "View", "Run", and "Help". The toolbar contains icons for file operations and execution controls. The main interface is divided into several sections:

- ROM Panel:** A list of memory addresses and their corresponding instructions. The first instruction at address 0 is "D=M".
- RAM Panel:** A list of memory addresses and their values. The first value at address 0 is 23773.
- Screen Display:** A large black rectangle representing the screen, with a small keyboard icon at the bottom.
- Keyboard Input:** A section labeled "D" with a text input field containing the value "0".
- ALU Unit:** A section labeled "ALU" with two input fields: "D Input" and "M/A Input", both containing "0". The "ALU output" field also contains "0".
- PC and A Registers:** At the bottom left, there are two input fields labeled "PC" and "A", both containing "0".

At the bottom of the window, a status bar indicates "Running...".